A Survey of Reinforcement Learning from Human Preferences

Summary

December 23, 2025

1 Problem Setup

Reinforcement Learning from Human Preferences (RLHF) has emerged as a powerful paradigm for training AI systems to align with human values and intentions. This comprehensive survey provides a systematic overview of the field, covering theoretical foundations, practical applications, and future research directions. The goal is to organize the rapidly growing literature on RLHF and provide guidance for researchers and practitioners.

2 Thesis

The central thesis is that RLHF represents a fundamental shift in how we train AI systems, moving away from explicit reward engineering toward learning from human judgment and preferences. The authors argue that RLHF has become essential for training safe and helpful AI systems, particularly large language models, but faces significant challenges that require ongoing research.

3 Key Concepts

Preference-Based Learning: Learning from human comparisons between different outcomes rather than explicit numerical rewards. This is more natural for humans and avoids the need for precise reward engineering.

Reward Model: A machine learning model trained to predict human preferences, serving as a proxy reward function for reinforcement learning training.

Preference Elicitation: The process of collecting human feedback, including pairwise comparisons, rankings, ratings, and various other forms of preference data.

Policy Optimization: Using reinforcement learning algorithms (typically PPO) to train policies that maximize the learned reward model while maintaining alignment with the original model.

Alignment Tax: The performance cost incurred when aligning models with human preferences, potentially reducing capabilities on some tasks.

4 Methods

The survey organizes RLHF research into several key areas:

1. **Feedback Collection Methods:** Different approaches to gathering human preferences, including pairwise comparisons, rankings, ratings, and active learning strategies to minimize feedback collection costs.

2. **Reward Learning Algorithms:** Various methods for training reward models from preference data, including supervised learning, Bayesian approaches, and inverse reinforcement learning techniques.
3. **Policy Optimization Techniques:** Different reinforcement learning approaches for optimizing policies using learned reward models, including PPO, policy gradients, and safe RL methods.
4. **Applications and Domains:** How RLHF is applied across different domains, including language models, robotics, recommendation systems, and game playing.
5. **Evaluation and Benchmarks:** Methods for evaluating RLHF systems and standardized benchmarks for comparing different approaches.

The authors provide comprehensive taxonomies and frameworks for understanding the relationships between different approaches.

5 Results

The survey identifies several key findings and trends:

Empirical Success:

- RLHF has become the standard approach for aligning large language models, leading to systems like ChatGPT and Claude.
- Applications extend beyond language to robotics, recommendation systems, and other domains.
- Performance improvements are consistent across different model sizes and architectures.

Theoretical Advances:

- Growing understanding of the theoretical properties of preference-based learning.
- Convergence guarantees for various RLHF algorithms under different assumptions.
- Analysis of sample complexity and feedback efficiency.

Practical Challenges:

- High cost of human feedback collection remains a major bottleneck.
- Reward hacking and misalignment continue to be significant concerns.
- Scaling to more capable systems presents new challenges.

Emerging Trends:

- Move toward more efficient feedback collection methods.
- Integration with other alignment techniques like constitutional AI.
- Development of more sophisticated reward modeling approaches.

6 Significance

This survey is significant for several reasons:

- **Comprehensive Overview:** Provides the first systematic organization of the rapidly growing RLHF literature.
- **Research Roadmap:** Identifies key open problems and research directions for the field.
- **Practical Guidance:** Offers insights for practitioners implementing RLHF systems.
- **Historical Context:** Traces the evolution of the field from early preference learning to modern large-scale applications.
- **Interdisciplinary Connections:** Highlights connections between RLHF and related fields like human-computer interaction, behavioral economics, and social choice theory.

The survey serves as both a reference for researchers and a guide for understanding the current state and future directions of RLHF research, which has become crucial for the development of safe and beneficial AI systems.